

Disproof of Conjecture 00000008556: the fixed-point chain of the free distributive lattice has length $n - 1$, not $\log p(n)$

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Abstract

Conjecture 00000008556 asserts three things about the free distributive lattice $\text{FD}(n)$ on n generators: its automorphism group is the symmetric group S_n ; its fixed-point lattice is “of partition-symmetric type”; and the maximal chain length of the fixed-point lattice is exactly $\log p(n)$, where $p(n)$ is the partition number. We refute the third clause, and hence the conjunction. The automorphisms of $\text{FD}(n)$ are the permutations of the generators, and the elements they fix are the n threshold functions “at least k of the n generators”. These form a chain of length $n - 1$, so the lengths for $n = 1, 2, 3$ are $0, 1, 2$, while $p(1), p(2), p(3) = 1, 2, 3$. The equation $1 = \log_b 2$ forces $b = 2$, and then $2 = \log_2 3$ is false. So no base b of the logarithm works, and the natural logarithm fails already at $n = 2$. The argument also covers the bounded lattice (with $0, 1$ adjoined), chain length counted in elements, rounded logarithms, and an “eventually” reading. A further arithmetic fact makes the failure independent of how the chain lengths are computed: no base b makes both $\log_b 2$ and $\log_b 3$ nonnegative integers. Everything that the disproof uses is formalized in Lean 4 (core library only): $\text{FD}(n)$ (as the $\wedge/\vee$ -closure of the projections and as the monotone functions), the S_n -action, lattice automorphisms, fixed points, chains, and the partition numbers.

1 The conjecture and its precise reading

Definition: A free distributive lattice is a lattice of ideals. *Conjecture:* The automorphism group of the free distributive lattice is the symmetric group; its fixed-point lattice is of partition-symmetric type; and the maximal chain length of the fixed-point lattice is exactly the logarithm of the partition number $p(n)$.

The Chinese version says the same. Here $\text{FD}(n)$ is the free distributive lattice on $n \geq 1$ generators, and n is also the argument of $p(n)$. The conjecture is the conjunction of

- (C1) $\text{Aut}(\text{FD}(n)) = S_n$, acting by permuting the generators;
- (C2) the fixed-point lattice is “of partition-symmetric type” (this is not defined; we do not need it);
- (C3) there is a logarithm $\log = \log_b$ (base $b > 0$, $b \neq 1$; for example $b = e$, 2 or 10) such that for every $n \geq 1$ the maximal chain length $\ell(n)$ of the fixed-point lattice $\text{Fix}(n)$ equals $\log_b p(n)$.

Here $\text{Fix}(n) = \{x \in \text{FD}(n) : \varphi(x) = x \text{ for all } \varphi \in \text{Aut}(\text{FD}(n))\}$ is the set of elements fixed by the automorphism group (“its fixed-point lattice”), with the order of $\text{FD}(n)$. The length of a chain $x_0 < x_1 < \dots < x_k$ is k , the number of steps. We also treat the convention that counts elements $(k+1)$.

We prove $\neg(\text{C1} \wedge \text{C2} \wedge \text{C3})$ for every meaning of (C2). If (C1) is false the conjunction is false. If (C1) holds, then $\text{Fix}(n)$ is the set of S_n -fixed elements, and (C3) fails for it. (C1) is in fact true (Remark 3), but the proof does not use this.

The definition “lattice of ideals”. By Birkhoff’s representation theorem, a finite distributive lattice is the lattice of order ideals (down-sets) of its poset of join-irreducibles. For $\text{FD}(n)$ this gives the usual descriptions below: monotone Boolean functions correspond to up-sets of $\{0, 1\}^n$, that is, to the complementary down-sets. Every reading of the definition therefore gives the same lattice $\text{FD}(n)$, and the definition plays no further role.

2 Definitions and basic facts

Points of $\{0, 1\}^n$ are identified with subsets $x \subseteq [n] = \{1, \dots, n\}$. A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is *monotone* if $x \subseteq y$ and $f(x) = 1$ imply $f(y) = 1$. Functions are ordered pointwise, with $\wedge, \vee$ computed pointwise. The projection π_i is $\pi_i(x) = [i \in x]$.

Proposition 1. *The sublattice of $\{0, 1\}^{\{0, 1\}^n}$ generated by $\pi_1, \dots, \pi_n$ is the set of nonconstant monotone functions. It is the free distributive lattice $\text{FD}(n)$. Adding the constants 0, 1 gives the free bounded distributive lattice, which consists of all monotone functions. The sizes are $|\text{FD}(n)| = 1, 4, 18, 166$ and $3, 6, 20, 168$ in the bounded case, for $n = 1, 2, 3, 4$.*

Proof. Projections are monotone with $\pi_i(\emptyset) = 0$ and $\pi_i([n]) = 1$, and these three properties are preserved by $\wedge$ and $\vee$. Conversely, let f be monotone and nonconstant. Then $f(\emptyset) = 0$, since otherwise monotonicity gives $f \equiv 1$. Hence $f = \bigvee_{x: f(x)=1} \bigwedge_{i \in x} \pi_i$, where every x in the join is nonempty and the join is nonempty. That this lattice is free is the classical normal form theorem for distributive lattices (two lattice polynomials agree in every distributive lattice if and only if they agree on $\{0, 1\}$). The sizes are the Dedekind numbers $3, 6, 20, 168$ minus the two constants. $\square$

S_n acts on functions by permuting variables: $(\sigma f)(x) = f(\sigma(x))$. Each σ preserves $\wedge, \vee$ and $\text{FD}(n)$, so it is a lattice automorphism.

Proposition 2. *The S_n -fixed elements of $\text{FD}(n)$ are the threshold functions $T_k(x) = [|x| \geq k]$ with $1 \leq k \leq n$; in the bounded lattice they are those with $0 \leq k \leq n+1$. They form the chain $T_n < T_{n-1} < \dots < T_1$, of length $n-1$ (length $n+1$ in the bounded case).*

Proof. If f is fixed by all σ then $f(x)$ depends only on $|x|$, because S_n is transitive on the k -subsets. Write $f(x) = g(|x|)$. Monotonicity makes g nondecreasing on $\{0, \dots, n\}$, so $g = [\cdot \geq k]$ for some $0 \leq k \leq n+1$. The constants are $k=0$ ($f \equiv 1$) and $k=n+1$ ($f \equiv 0$). Conversely every T_k is symmetric and monotone. Since $T_{k+1} \leq T_k$, with strict inequality, the fixed set is totally ordered with n (respectively $n+2$) elements. $\square$

Remark 3 (Clause (C1) is true). For a finite distributive lattice L , the automorphisms of L correspond to the automorphisms of its poset $J(L)$ of join-irreducibles (Birkhoff duality). For $\text{FD}(n)$, J consists of the elements $\bigwedge_{i \in S} \pi_i$ with $\emptyset \neq S \subsetneq [n]$ (all $S \subseteq [n]$ in the bounded case), ordered by reverse inclusion. The singletons $\{i\}$ are the minimal elements of $J \setminus \{\emptyset\}$ under inclusion ($\emptyset$, present only in the bounded case, is the extreme element and is fixed). So an automorphism φ of

J permutes the singletons, say $\varphi(\{i\}) = \{\sigma(i)\}$. Since $\{i\} \subseteq S \iff \varphi(\{i\}) \subseteq \varphi(S)$ and every S is the union of the singletons it contains, $\varphi(S) = \sigma(S)$. Hence $\text{Aut}(\text{FD}(n)) \cong S_n$, acting by permuting the generators. The script `verify.py` confirms this by brute-force backtracking over all lattice automorphisms for $n \leq 3$, and over all automorphisms of J for $n = 4$. Consequently $\text{Fix}(n)$ is exactly the chain of Proposition 2.

Partition numbers. $p(n)$ is the number of nonincreasing sequences of positive integers with sum n : $p(1), \dots, p(5) = 1, 2, 3, 5, 7$.

3 The counterexample

For real $b > 0$ with $b \neq 1$, real $x > 0$ and real ℓ , by the definition of the logarithm,

$$\ell = \log_b x \iff b^\ell = x. \quad (1)$$

Chain lengths are natural numbers, so only powers b^ℓ with $\ell \in \mathbb{N}$ occur.

Theorem 4. *Let $\ell(n)$ be the maximal chain length of the S_n -fixed lattice of $\text{FD}(n)$. There is no base b with $\ell(n) = \log_b p(n)$ for all $n \geq 1$. The same holds for the bounded lattice, and for chain length counted in elements. Consequently conjecture 00000008556 is false.*

Proof. By Proposition 2, the lengths at $n = 1, 2, 3$ are $s, s+1, s+2$, where $s = 0$ (free lattice, steps), $s = 1$ (free, elements), $s = 2$ (bounded, steps) or $s = 3$ (bounded, elements). Also $(p(1), p(2), p(3)) = (1, 2, 3)$. By (1), the clause gives $b^s = 1$, $b^{s+1} = 2$ and $b^{s+2} = 3$. The first two give $b = b^{s+1}/b^s = 2$. Then $b^{s+2} = b^{s+1} \cdot b = 4 \neq 3$, a contradiction. In particular, for the natural logarithm and the free lattice: $\ell(2) = 1 \neq \ln 2$. For the conjecture: if (C1) fails, the conjunction fails. If (C1) holds, then $\text{Fix}(n)$ equals the S_n -fixed lattice, and (C3) fails by the above. $\square$

The only algebra used is $b \cdot 1 = b$, associativity, and $2 \cdot 2 = 4 \neq 3$ for numbers. In Lean the base b therefore ranges over an arbitrary “number system”: any type with an associative unital multiplication and an injective, multiplicative map from $\mathbb{N}$. Examples are $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

4 Robustness

Free or bounded lattice; steps or elements. All four combinations are covered by Theorem 4 and by the Lean theorem.

Any base. The base b is arbitrary: e , 2, 10, or even $0 < b < 1$. The value $b = 2$ is forced by $n \leq 2$, and in fact the clause *does* hold for $n = 1, 2$ with $b = 2$ (Lean: `clause_holds_n_le_2`). The failure is at $n = 3$, where $\log_2 3 \approx 1.585 \neq 2$.

Any chain-length convention and any “fixed-point lattice”.

Proposition 5. *For a real $b > 0$ and natural numbers L_2, L_3 , it is impossible that $L_2 = \log_b p(2)$ and $L_3 = \log_b p(3)$.*

Proof. Otherwise $b^{L_2} = 2$ and $b^{L_3} = 3$, so $2^{L_3} = b^{L_2 L_3} = 3^{L_2}$. By parity $L_3 = 0$, so $3^{L_2} = 1$ and $L_2 = 0$. Then $b^0 = 1 \neq 2$. $\square$

So, whatever is meant by the fixed-point lattice, the clause fails at $n = 2, 3$, as long as chain lengths are integers. This includes the fixed lattice of a single automorphism, which is not what “its” (the group’s) says. For example `verify.py` prints the lengths 2, 1 for $n = 2$ and 6, 4, 2 for $n = 3$ for the conjugacy classes of S_2 and S_3 (identity, transposition, 3-cycle). Proposition 5 is proved in Lean as `no_lengths_fit`. The same applies to the poset of S_n -orbits on $\text{FD}(n)$, or to any quotient lattice, since its chain lengths are integers too. Since the proposition uses only $n = 2, 3$, it also covers readings that exclude $n = 1$.

Rounded logarithms. “Exactly” excludes rounding. Even so, $\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ and nearest-integer rounding of $\log_b p(n)$ for $b \in \{2, e, 10\}$ all fail to equal $n - 1$ at some $n \leq 5$ (`verify.py` prints the first failure). For example $\lceil \log_2 p(5) \rceil = \lceil \log_2 7 \rceil = 3 \neq 4$.

“Eventually” or asymptotic readings. For every base $b > 1$ the clause fails for all large n , even up to a bounded error. For $0 < x < 1$ and every n ,

$$p(n) x^n \leq \sum_{m \geq 0} p(m) x^m = \prod_{k \geq 1} \frac{1}{1 - x^k} =: F(x) < \infty,$$

so $p(n)^{1/n} \leq x^{-1} F(x)^{1/n}$ and $\limsup_n p(n)^{1/n} \leq 1/x$. Letting $x \rightarrow 1$ gives $p(n)^{1/n} \rightarrow 1$, that is, $\log_b p(n) = o(n)$. Hence $|(n - 1) - \log_b p(n)| \rightarrow \infty$, and the identity, rounded or not, can hold for only finitely many n . For $0 < b < 1$ the logarithm is negative for $n \geq 2$. This paragraph is not formalized; the exact statement (Theorem 4) is.

5 Lean formalization

The project `lean4/` uses Lean 4.19.0 with the core library only (no Mathlib). All objects are defined from scratch in `Main.lean`. A Boolean function on n variables is a truth table encoded as a natural number $f < 2^{2^n}$: bit m is the value at the point m , and coordinate i of m is bit i of m .

- **Gen n bnd**: the inductive closure of the projections `proj n i` under `&&&` and `|||`, with 0 and 1 when `bnd = true`. **FDset n bnd**: monotone and, unless bounded, nonconstant truth tables. `fd_spec` proves, for $1 \leq n \leq 3$, that these coincide and equal an explicit list. The proof combines a general soundness lemma for the closure computation `closureL` with kernel checks over all 2^{2^n} truth tables. `fd_sizes` gives 1, 4, 18 and 3, 6, 20.
- **IsPerm, act**: permutations as duplicate-free lists and their action on variables. **Fixed**: the S_n -fixed elements. `fixed_iff_thr`: for $1 \leq n \leq 3$, the fixed elements are exactly the thresholds `threshold n k`, with k in the admissible range.
- **IsAut**: a lattice automorphism (a bijection of $\text{FD}(n)$ preserving $\wedge, \vee$). **AutIsSym**: clause (C1). **FixedAut**: elements fixed by every automorphism. `act_isAut`: each permutation is an automorphism. `fixedAut_iff`: under (C1), `FixedAut = Fixed`.
- **MaxChainLen P k**: some chain in P has $k + 1$ elements and none has more. `maxChain_fixed`: lengths 0, 1, 2 (free) and 2, 3, 4 (bounded) for $n = 1, 2, 3$. The upper bound uses a pigeonhole lemma (`nodup_len_le`).
- **IsPartition, partitionsL, p**: `mem_partitionsL` proves for every n that `partitionsL n` lists exactly the partitions of n . `p_values`: $p(1..5) = 1, 2, 3, 5, 7$, and the lists for $n \leq 3$ have no duplicates.

- `NumSys`, `IsLog R b x y :  $\iff b^y = x$` (equation (1)), and `LogClause R bnd elems: clause (C3)` for the S_n -fixed lattice.

Main results:

- `logClause_false:  $\neg$ LogClause R bnd elems`, for all R , `bnd`, `elems`;
- `conjecture_00000008556_false:  $\neg$ Conjecture R C2 bnd elems`, where `Conjecture` is $(C1) \wedge (C2) \wedge (C3 \text{ for FixedAut})$, for an arbitrary predicate `C2`;
- `no_lengths_fit`: Proposition 5;
- non-vacuity: `natSys`, `intSys` are number systems, `act_isAut` shows that `IsAut` is satisfiable, and `clause_holds_n_le_2` shows that the clause holds for $n \leq 2$ with $b = 2$.

There is no `sorry`, no `native_decide` and no added axiom. `#print axioms` reports at most `propext` and `Quot.sound`.

Limitations. Core Lean has no real numbers. The theorems are stated for an arbitrary `NumSys`; $\mathbb{R}$ satisfies its axioms, and by (1) the real-number clause is an instance of `LogClause`. Propositions 1 and 2 for general n , the truth of (C1) (Remark 3) and the asymptotic paragraph are proved in this report only; Lean covers $n \leq 3$, which is all the disproof needs. Clause (C2) is not formalized; it enters only as an arbitrary predicate.

6 Independent check

`verify.py` (Python 3 standard library) represents a function by its set of true points and uses a different method. For $n \leq 4$ and both lattices it computes $\text{FD}(n)$ as a closure and compares it with the set of monotone functions for $n \leq 3$. It computes $\text{Aut}(\text{FD}(n))$ without assumptions: by backtracking over lattice automorphisms for $n \leq 3$, and over automorphisms of the join-irreducible poset for $n = 4$. It then finds the fixed points of the whole group (the thresholds) and the longest chain. It also computes $p(n)$ by Euler’s pentagonal recurrence, cross-checked by direct enumeration, and checks every reading of Section 4.

7 Reproduction

```
cd lean4 && lake build
cd .. && python3 verify.py
pdflatex report.tex && pdflatex report.tex
```